

KILIMANJARO INTL, Tanzania, United Republic of

WMO: 637910

Lat: 3.429S

Lon: 37.074E

Elev: 894

StdP: 91.04

Time Zone: 3.00 (E03)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.4	14.8	8.5	7.7	28.5	9.8	8.4	26.7	8.3	25.3	7.5	24.9	0.4	330	0.329

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.3	34.8	19.9	33.9	20.0	33.1	20.3	22.7	28.6	22.3	28.3	22.0	28.1	5.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	17.7	24.0	21.0	17.5	23.6	20.6	17.0	23.3	72.2	28.7	70.4	28.4	69.0	27.8	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.3	9.3	8.2	DB	10.8	36.6	1.0	1.2	10.0	37.4	9.4	38.1	8.8	38.7	8.1	39.6	(4)
(5)			WB	9.6	25.0	1.7	1.1	8.4	25.8	7.4	26.4	6.5	27.1	5.3	27.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	24.1	26.1	26.4	26.4	25.0	22.9	21.4	21.0	21.8	22.9	24.6	25.6	25.7	(6)
(7)		DBStd	2.33	1.16	1.15	1.37	1.43	1.25	1.38	1.30	1.22	1.35	1.41	1.11	1.10	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0	(9)
(10)		CDD10.0	5158	499	459	510	450	401	342	341	364	386	452	467	486	(10)
(11)		CDD18.3	2116	241	226	252	200	143	92	83	106	136	194	217	228	(11)
(12)		CDH23.3	16915	2453	2308	2332	1304	591	396	408	618	1067	1644	1798	1997	(12)
(13)		CDH26.7	6604	1168	1111	1080	433	90	37	37	94	316	658	743	838	(13)
(14)	Wind	WSAvg	3.1	4.3	4.4	3.7	2.5	1.8	1.7	1.9	2.5	3.2	3.7	3.9	4.0	(14)
(15)	Precipitation	PrecAvg	931	45	41	104	278	186	39	26	16	35	71	69	(15)	
(16)		PrecMax	1371	183	153	285	498	299	119	49	79	114	159	217	187	(16)
(17)		PrecMin	630	0	0	24	89	41	16	8	4	3	2	9	0	(17)
(18)		PrecStd	217	46	36	57	113	60	19	8	16	22	38	45	46	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.2	35.7	35.5	33.2	30.2	28.9	29.0	30.1	31.9	33.4	33.6	34.1	(19)
(20)			MCWB	19.3	20.3	20.7	21.0	20.7	19.7	18.5	18.6	18.9	19.2	20.0	19.6	(20)
(21)		2%	DB	34.2	34.8	34.2	31.9	28.9	27.7	27.5	28.9	30.8	32.2	32.8	33.1	(21)
(22)			MCWB	19.7	20.0	20.4	21.0	20.5	19.2	18.1	18.2	18.8	19.1	20.0	20.1	(22)
(23)		5%	DB	33.2	33.6	33.2	30.6	27.8	26.9	26.8	27.9	29.9	31.2	31.9	32.1	(23)
(24)			MCWB	20.1	20.2	20.6	20.9	20.4	18.9	18.0	17.9	18.4	19.2	20.0	20.4	(24)
(25)		10%	DB	32.3	32.6	32.1	29.2	26.8	25.8	25.8	26.9	28.9	30.4	30.9	31.1	(25)
(26)			MCWB	20.3	20.3	20.6	21.2	20.3	18.7	17.7	17.9	18.2	19.2	20.2	20.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.7	22.6	23.1	23.2	22.2	20.7	20.1	20.1	20.7	22.1	22.9	23.2	(27)
(28)			MCD B	29.6	29.8	29.9	27.9	25.8	25.6	26.3	27.2	28.1	28.0	28.9	28.7	(28)
(29)		2%	WB	22.1	22.1	22.4	22.6	21.7	20.2	19.2	19.3	19.9	21.1	22.1	22.4	(29)
(30)			MCD B	29.4	29.6	29.2	27.4	25.5	25.0	25.0	25.9	27.2	26.8	28.2	28.2	(30)
(31)		5%	WB	21.6	21.6	22.0	22.1	21.2	19.7	18.7	18.8	19.4	20.6	21.5	21.8	(31)
(32)			MCD B	28.9	29.0	28.8	26.9	25.1	24.4	24.3	25.1	26.3	26.4	27.6	27.9	(32)
(33)		10%	WB	21.2	21.2	21.6	21.7	20.8	19.2	18.2	18.3	18.9	20.2	21.1	21.4	(33)
(34)			MCD B	28.4	28.4	28.1	26.4	24.6	23.6	23.7	24.6	25.7	26.2	27.1	27.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.4	13.3	12.0	9.1	7.9	9.4	10.4	10.9	12.5	12.2	11.6	11.9	(35)
(36)			MCD BR	15.0	14.9	13.6	10.9	9.4	10.9	11.9	12.6	14.4	13.6	12.8	13.3	(36)
(37)			MCWBR	3.9	3.8	3.3	2.8	3.0	4.1	4.6	4.2	4.7	3.7	3.3	3.6	(37)
(38)		5% WB	MCD BR	12.5	12.9	11.4	9.1	7.7	9.4	10.4	10.9	12.3	11.6	11.1	11.2	(38)
(39)			MCWBR	3.9	3.7	3.2	3.1	3.0	3.9	4.4	4.0	4.5	3.5	3.3	3.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.452	0.489	0.451	0.376	0.369	0.414	0.439	0.428	0.409	0.406	0.403	0.434	(40)	
(41)		taud	2.241	2.115	2.262	2.513	2.511	2.337	2.223	2.259	2.317	2.340	2.390	2.319	(41)	
(42)		Ebn at Noon	885	854	877	922	903	846	830	862	902	917	925	897	(42)	
(43)		Edh at Noon	147	168	143	108	104	120	137	137	134	132	125	135	(43)	
(44)	All-Sky Solar Radiation	RadAvg	6.31	6.49	6.13	5.29	4.59	4.37	4.49	4.94	5.84	6.11	5.80	5.94	(44)	
(45)		RadStd	0.41	0.38	0.44	0.31	0.21	0.21	0.39	0.35	0.34	0.33	0.29	0.34	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page